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# Comparison of Palaeomagnetic Results for Selected Rocks of Great Britain and North America

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RECENTLY it has been possible to compare palaeomagnetic measurements from rocks on either side of the North Atlantic and from such comparisons to deduce that relative movements between North America and Europe have taken place. This paper proposes to describe briefly three periods of geologic time from which sufficient palaeomagnetic evidence has been collected to warrant making certain conclusions. These three periods are late Pre-Cambrian, Permian and Upper Triassic.

The author has measured an extensive suite of Keweenawan rocks (Pre-Cambrian) which were collected from around Lake Superior. The sub-divisions of the Keweenawan, from which samples were measured, are Portage Lake, Copper Harbour, Freda and Nonesuch, Jacobsville and Chequamegon. The Portage Lake, which has been generally classified as Middle Keweenawan, consists of an enormous thickness of lava flows, principally basalt and andesite, which are interbedded with some thin layers of conglomerate and sandstone. The Copper Harbour, which is considered as Upper Keweenawan, as are all the rest of the above mentioned formations, is mainly a very coarse conglomerate, consisting principally of boulders of rhyolite and other more acidic rocks in a sandy matrix. A subordinate amount of the boulders are basalt and andesite, identical with the underlying lavas. There are some lenses of medium to coarse grained reddish sandstone interbedded with the conglomerate. Furthermore, there are some basaltic and andesitic lavas in the Copper Harbour. There is a sharp lithological break at the top of the Copper Harbour, where coarse conglomerates pass up into fine grained sandstone, siltstone and shale within a few feet. This sandstone and shale is the Nonesuch, which, with the overlying Freda, consists of fine grained reddish or purplish grey sandstone, siltstone and shale. The Jacobsville is mainly a fine to medium grained sandstone, brick red in colour except for many white spots where evidently reduction of red haematite has taken place. The Chequamegon is a medium to coarse grained sandstone, which is the stratigraphically highest Keweenawan formation to be found. The palaeomagnetic results for these formations are plotted on a stereographic polar projection in figs. 1–5. The conventional notation, except where stated otherwise, is that full circles are north poles on the lower hemisphere and open circles are north poles on the upper hemisphere.

The magnetic stability of these rocks may be shown in a number of ways. The fact that except for the Chequamegon the remanent directions of magnetization are quite far from the present field is in itself a good

indication of fairly strong stability. Pebbles of basaltic and andesitic lava were collected from the Copper Harbour. Their measured directions are plotted in fig. 6. The randomness of this plot indicates that the lavas have been magnetically stable since Copper Harbour time. In the Freda and Nonesuch it is possible to find rocks some of which are flat lying and others practically vertical. Some of these rocks are plotted in figs. 7(*a*) and 7(*b*) to show their remanent direction without, and with, correction for geologic dip. The crosses on fig. 7(*a*) are the bedding poles plotted on the upper hemisphere. Thermal decay curves have been obtained for selected rocks from the Keweenawan, and they all exhibit a thermal decay similar to that of thermo-remanent magnetization with no change in the direction of the vector.

Samples of Permian rocks have been collected from the Mauchline basin of Scotland. In this locality are found 500 ft of olivine-basalt lava overlain by approximately 2000 ft of red and orange desert sandstone. Samples of both lithologies have been measured and are plotted in figs. 8(*a*) and 8(*b*). Although the sediments are more scattered it can be seen at a glance that they give a result quite consistent with the lavas.

| Formation | Declination *D* | Dip *I* | No. of Specimens *N* | $\alpha$† | Pole position | |
|---|---|---|---|---|---|---|
| Portage Lake | 282° E | +41° | 31 | 4° | 170° W | 25° N |
| Copper Harbour | 294° E | +32° | 25 | 7° | 176° E | 30° N |
| Freda and Nonesuch | 285° E | − 1° | 68 | 3° | 169° E | 9° N |
| Jacobsville | 250° E | −11° | 15 | 13° | 170° W | 14° S |
| Chequamegon | 30° E | +74° | 15 | 6° | 133° E | 69° S |
| Mauchline lavas | 180° E | − 4° | 34 | 8° | 175° E | 36° N |
| Mauchline sandstones | 187° E | − 6° | 26 | 12° | 163° E | 37° N |
| Triassic | 12° E | +14° | 32 | 3° | 91° E | 41° N |

† Radius of 95% circle of confidence calculated by Fisher's Method (1953).

Finally samples of Triassic rocks from the Connecticut Valley in the State of Connecticut were collected and measured in order to compare the Triassic of America with the Triassic of Great Britain, which has been so extensively measured by Clegg *et al.* (1954). These results are plotted in fig. 9. The fact that there is a reversed group would indicate that these rocks do not have very much isothermal remanent magnetization superimposed on them and are therefore stable.

It is possible to compute for each mean direction of magnetization found in the above mentioned rocks a corresponding position of the magnetic pole. This computation assumes that secular variation and random errors will be eliminated in the process of meaning and that the earth's magnetic field is essentially a dipole. In the table the mean

directions, the number of samples, the 95% circle of confidence and the mean pole positions, are listed.

These pole positions and those found by other workers have been plotted for late Pre-Cambrian time in figs. 10(*a*) and 10(*b*). In fig. 10(*a*) the European and North American continents are assumed to have been in the same relative positions in the late Pre-Cambrian as they are today. The pole positions are marked by letters ; A for the Adirondack results of Buddington and Balsley (1954) ; G for dykes from Michigan measured by Graham (1953) ; H for the Hakatai shale measured by Runcorn (1956) ; P for the Portage Lake ; C for the Copper Harbour ; F for the Freda and Nonesuch ; J for the Jacobsville ; B for the Chequamegon ; D and T for the lower and upper Torridonian of Scotland measured by Creer *et al.* (1954). Figure 10(*b*) shows the relative positions of the poles if it is assumed that America has moved 45° westward in longitude from Europe since the late Pre-Cambrian. This shows that the results from Europe and North America agree much more closely if one assumes that such a shift has taken place.

In figs. 11(*a*) and 11(*b*) the same kind of plotting of pole positions for Permian time has been done. In fig. 10(*a*) no continental drift has been considered while in fig. 10(*b*) a 45° drift westwards in longitude of America from Europe is postulated. Once again the results are much more in accordance with the idea that continental drift of such a magnitude has taken place since Permian times. The various pole positions are marked as follows :—

L for Mauchline lavas ; S for Mauchline sandstones ; E for the Exeter Traps measured by Creer *et al.* (1954) ; R for the Supai of southwestern America measured by Runcorn (1956) ; D for the Supai measured by Doell (1955) ; G for the Supai measured by Graham (1955).

The pole positions for Triassic time are plotted in figs. 12(*a*) and 12(*b*). The pole marked C is that of the Connecticut Triassic while that marked T is the pole found by Clegg *et al.* (1954). The same relative shift of America with respect to Europe of 45° west in longitude brings the results in good agreement.

These results indicate that America may well have drifted westwards from Europe by the rather large amount of 45° since Triassic times. This is in good agreement with the suggestions which were made by Alfred Wegener at the beginning of this century and which have been, up to now, viewed with great scepticism by most geologists.

## References


Balsley, J. R., and Buddington, A. F., 1954, *Tech. Comm. Palaeomagnetism*, Xth. Assembly, A.T.M.E., I.U.G.G., Rome.

Clegg, J. A., Almond, M., and Stubbs, P. H. S., 1954, *Phil. Mag.*, **45**, 583.

Creer, K., Irving, E., and Runcorn, S. K., 1954, *J. Geomag. Geoelect.*, **6**, 163.

Doell, R. R., 1955, *Nature*, **176**, 1167.

Fisher, R. A., 1953, *Proc. Roy. Soc.* A, **217**, 295.

Graham, J. W., 1953, *J. Geophys. Res.*, **58**, 243 ; 1955, *Ibid.*, **60**, 329.

Runcorn, S. K., 1956, *Bull. Geol. Soc. Amer.*, **67**, 301.

Fig. 1

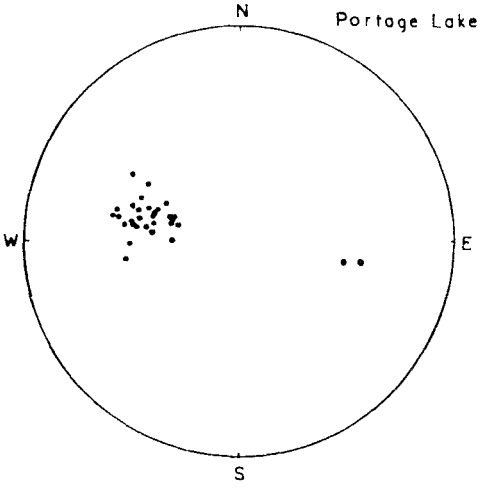



Fig. 2

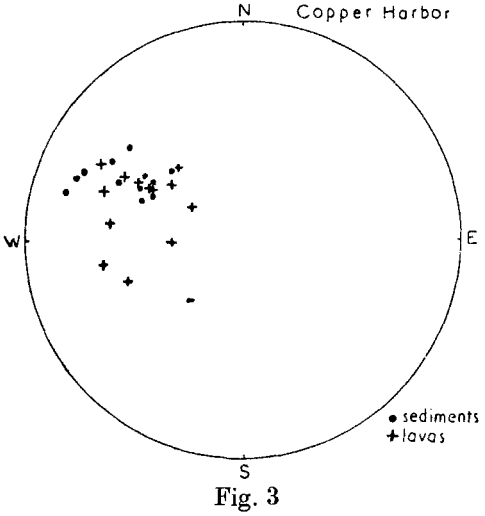
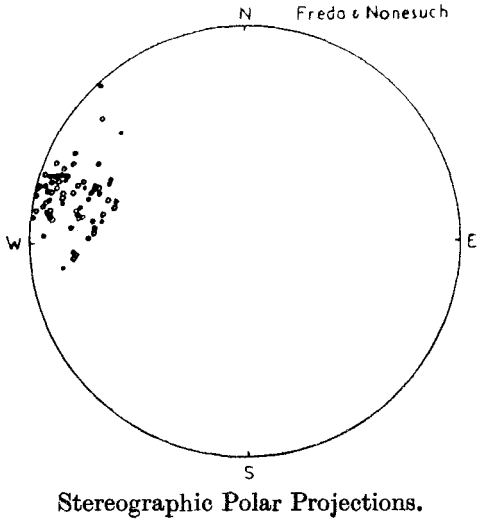



Fig. 3




Stereographic Polar Projections.

**Fig. 4**

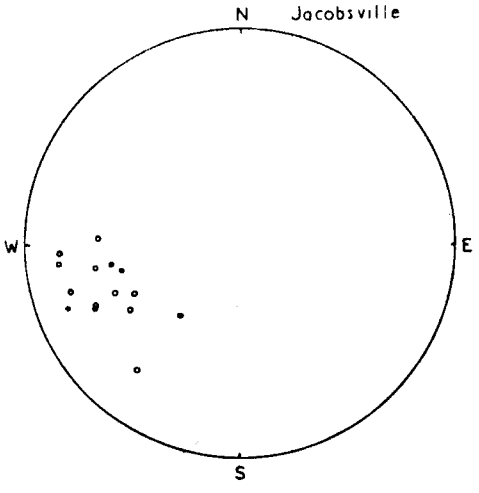



**Fig. 5**

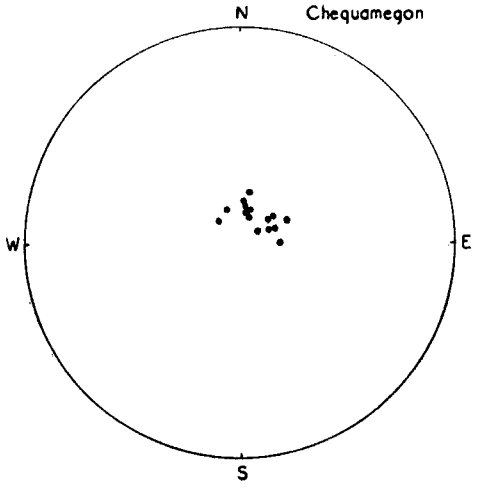
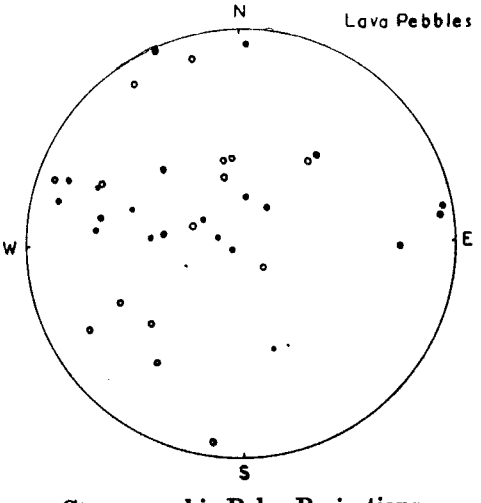



**Fig. 6**




**Stereographic Polar Projections.**

Fig. 7(*a*)

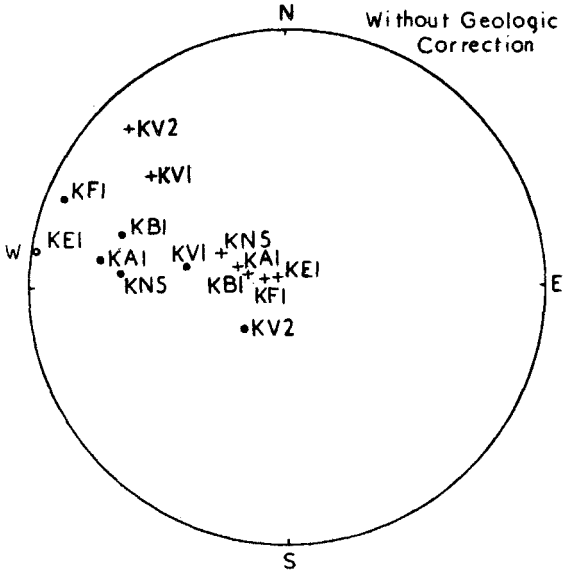
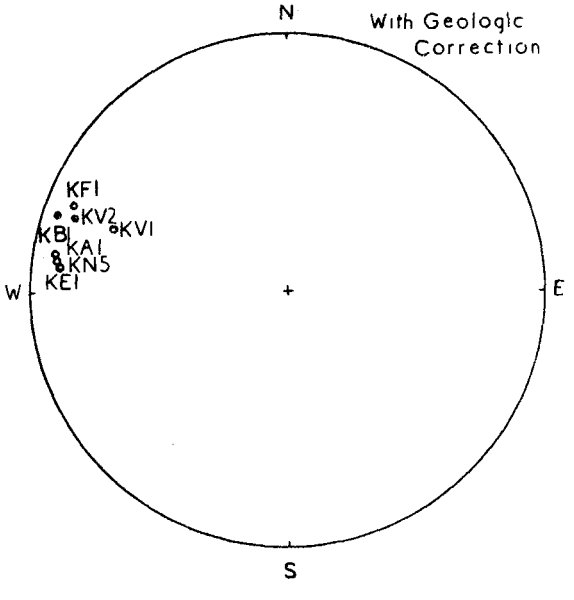



Fig. 7(*b*)




Stereographic Polar Projections.

Fig. 8(*a*)

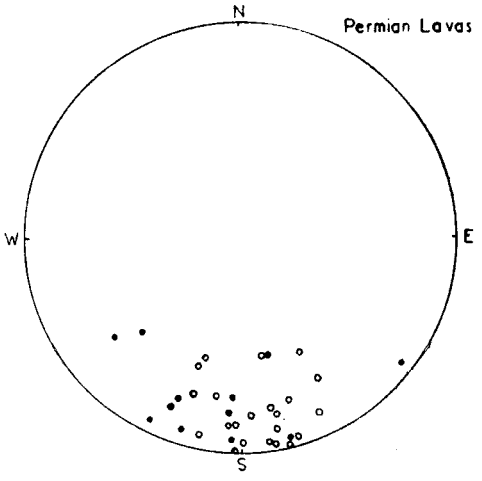



Fig. 8(*b*)

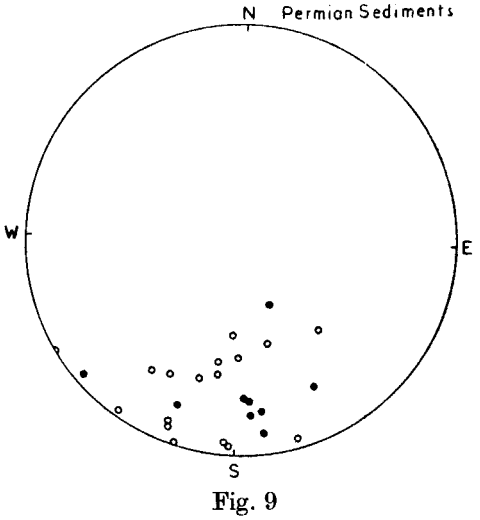
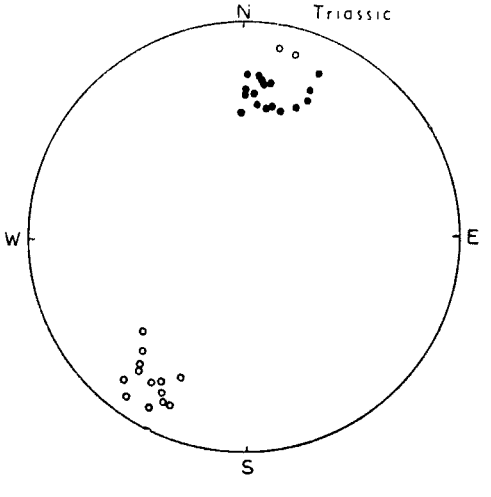



Fig. 9




Stereographic Polar Projections.

Fig. 10(*a*)

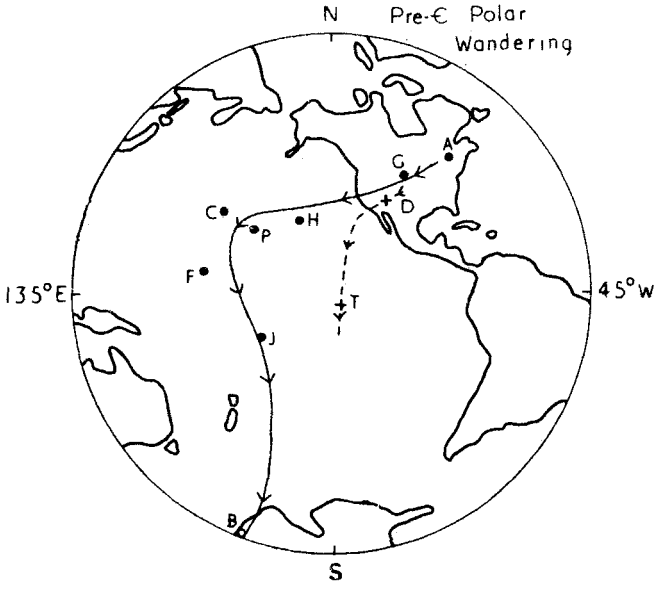



Fig. 10(*b*)

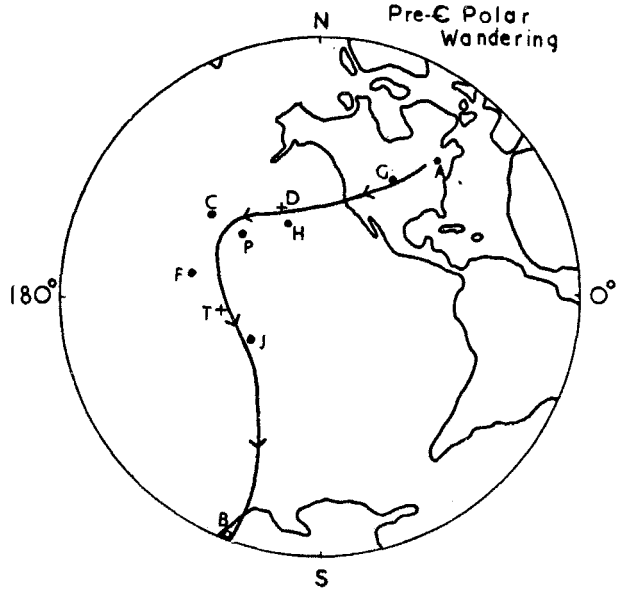


Fig. 11(*a*)

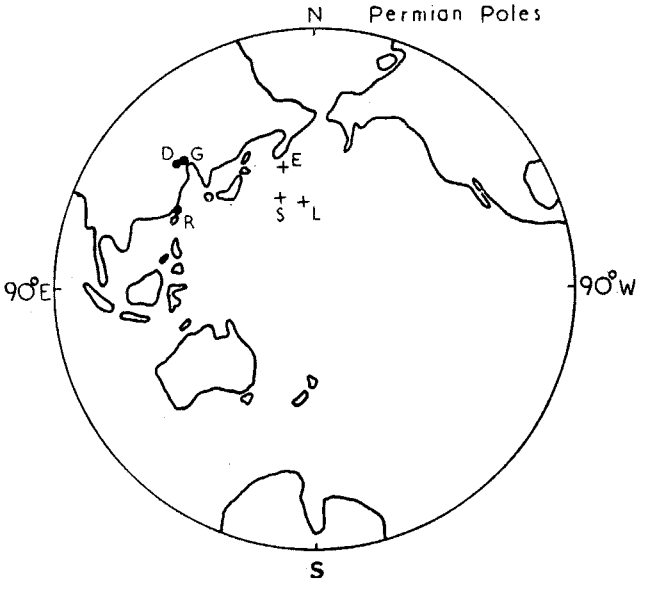



Fig. 11(*b*)

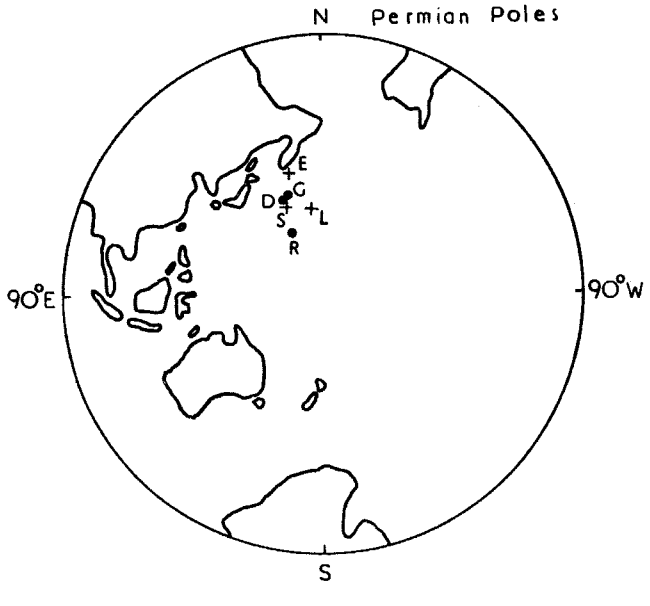


Fig. 12(*a*)

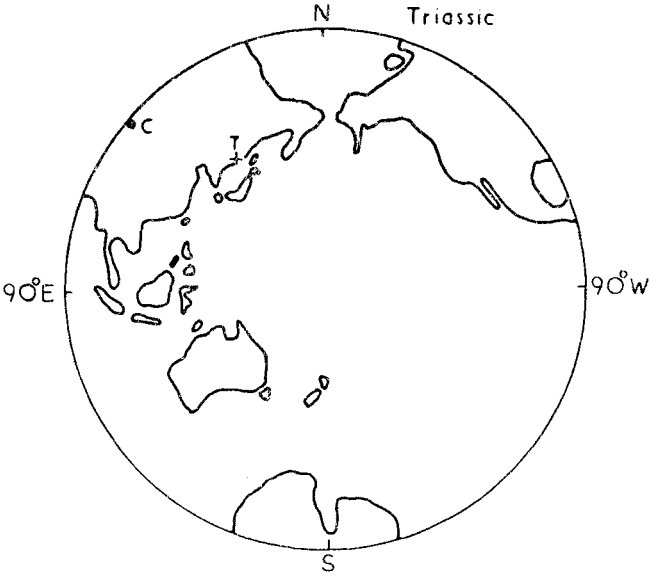



Fig. 12(*b*)

